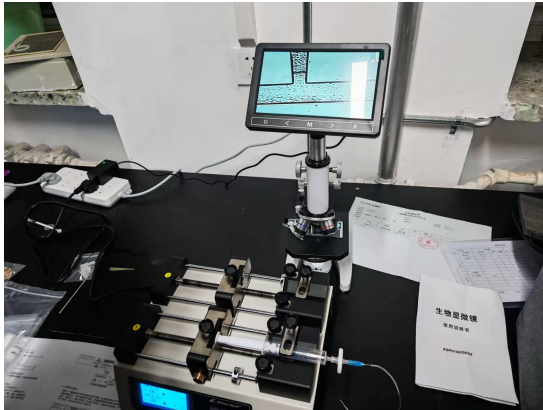


Lab Notebook (Microfluidic Platform)

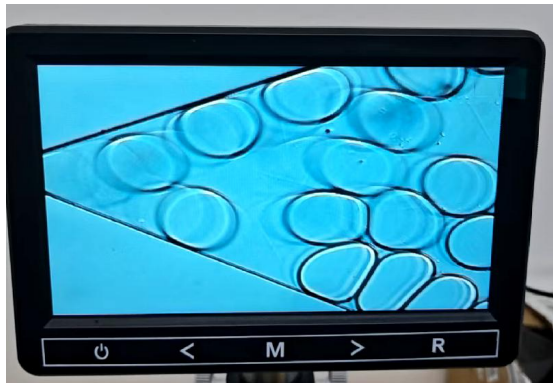
May

Building the Droplet-Generation Module

Completed the construction of the droplet-generation module: the cell suspension is injected into the chip by a syringe pump; after droplet generation, signal detection and droplet sorting are performed in sequence.



The platform can currently generate droplets, but the droplet size is non-uniform, which is suspected to originate from the working principle of the peristaltic pump.

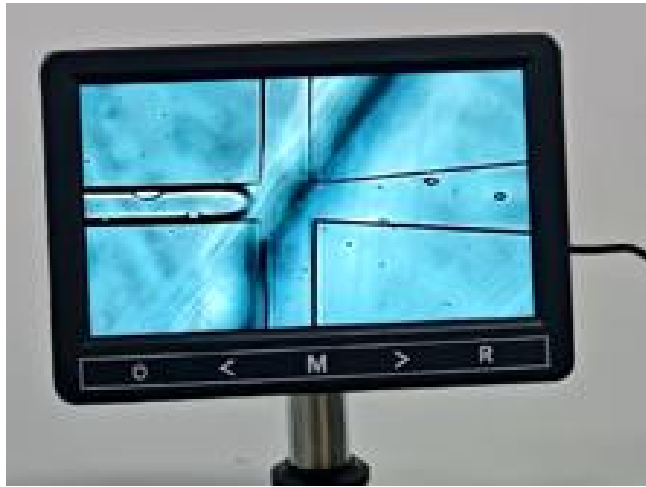


Video 1 — Droplet generation process

In terms of hardware, a 5 mL oil-phase container and a 100 μ L microsyringe are now in place; after replacement, the continuity of the syringe pump's fluid output improved. Improper operation during experiments was also found to readily cause microfluidic-chip clogging.

Initial Droplet Generation

Droplets were successfully and stably generated at a slow rate using LB/bacterial suspension, with a water-to-oil ratio of 1:10 and an oil-phase flow rate of $20 \mu\text{L/min}$.



Video 2 — Slow droplet generation (LB/bacterial suspension)

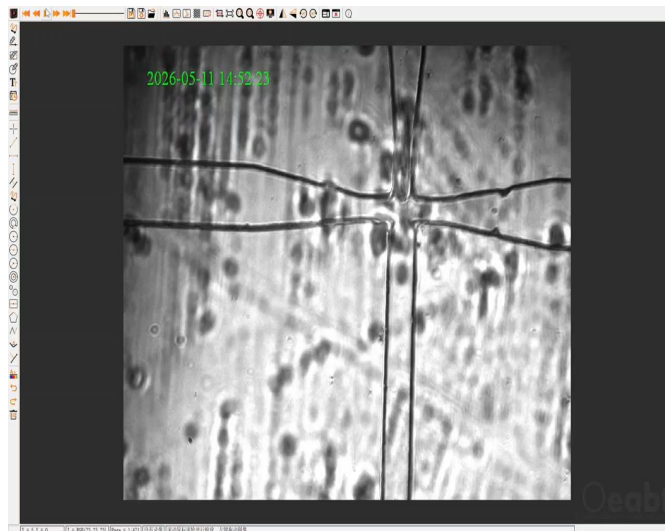
The chip is easily clogged by impurities.



Single-cell-encapsulated droplets produce a weak signal and are difficult to observe; incubation is planned to address this.

Stable Droplet Generation and Construction of the Optical Breadboard

Droplets are generated stably, uniformly, and in large quantities.



Video 3 — Stable, uniform droplet generation

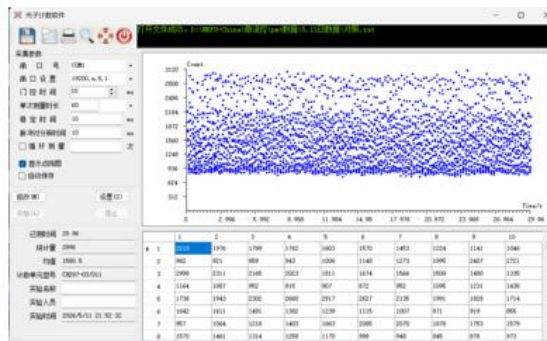
Overnight incubation loss $\leq 10\%$, and $\leq 5\%$ when simulating the complete process.



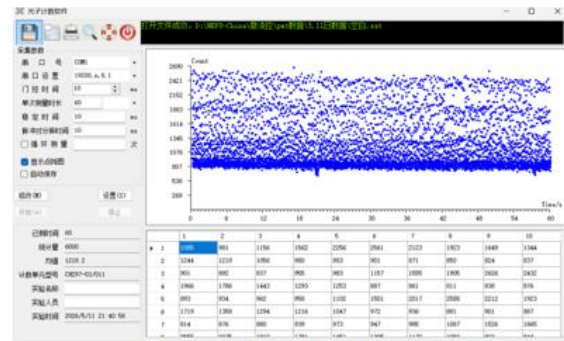
Video 4 — Overnight droplet incubation

The optical breadboard was built; fluorescence detection intensity was verified separately using a blank chip, pure oil phase, and droplets.

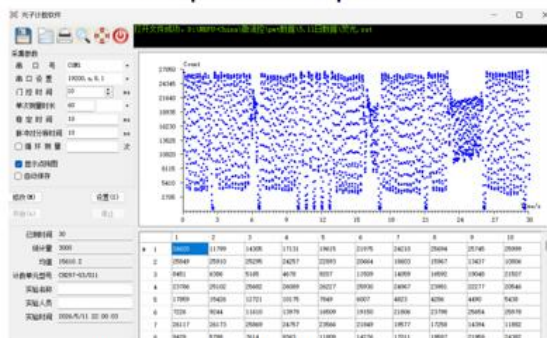
This demonstrated that the optical breadboard can distinguish fluorescent proteins from the oil phase.



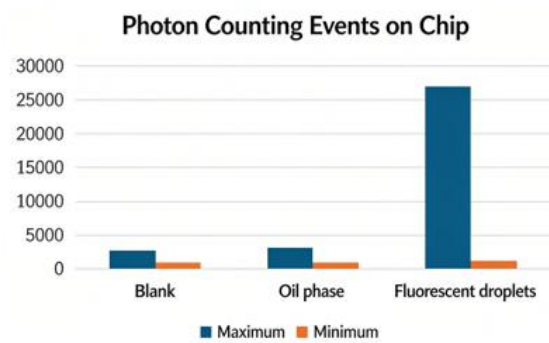
Chip with oil phase



Chip only (blank control)



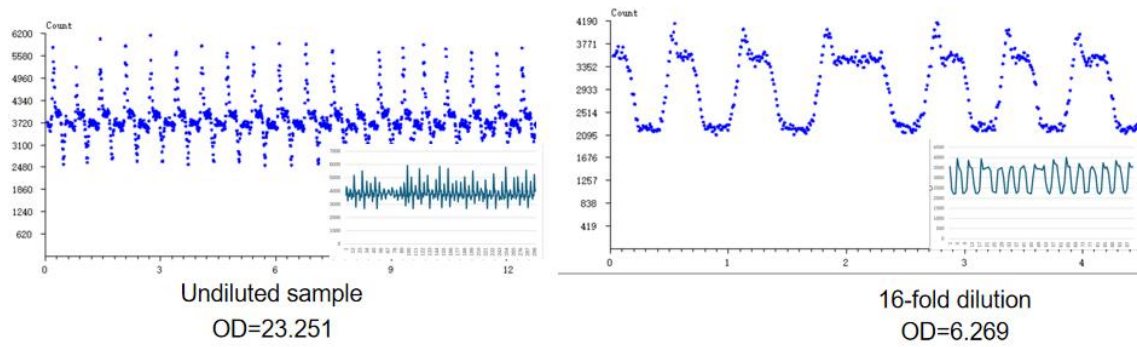
Chip with oil phase + droplets



Fluorescence-Detection Validation After Sample Incubation

Completed the incubation and preliminary fluorescence-detection validation of droplet samples: the original suspension and 16-fold-diluted droplets were incubated separately, yielding OD values of 6.269 and 23.251. PMT detection confirmed that the fluorescence-intensity

difference could still be clearly distinguished even at the lower OD, indicating that only a short incubation is needed before subsequent fluorescence experiments to validate the hardware system.



All videos referenced in this document are available for download at the bottom of our team's Notebook page.